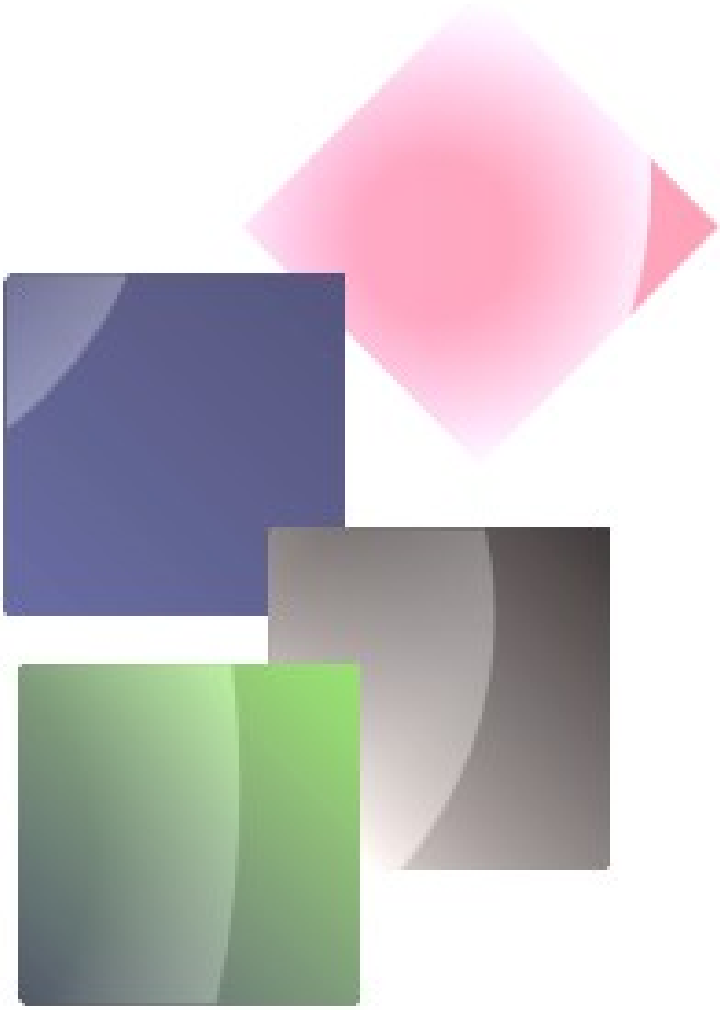
SyberPay Specifications Document (Merchants)



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Document Control

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1. Document Purpose

This document is intended to serve as a full reference guide for any party desiring to integrate their web applications and/or websites with SyberPay the payment aggregator. The document lists all the details of the various services provided by SyberPay, the flows of the messages and these messages' contents in details.

2. Introduction

SyberPay is the first CBoS certified payment aggregator in Sudan. It is a platform that aggregates several payment options and exposes them as simpler interfaces for integration. SyberPay will process the ePayments made at any merchant's website. Currently SyberPay platform supports a verity of payment means including debit cards & cash cards (ePurse Cards).

Figure 1 illustrates the most basic work flow for a customer paying using SyberPay. Initially the merchant integrates his/her website with SyberPay. After the registration the merchant can offer SyberPay as a payment option for customers using the website. Once a customer chooses to pay using SyberPay, the customer is redirected to SyberPay portal where he/she will be prompted to enter the card details. Then SyberPay will handle all of the payment operation on behalf of the merchant website and return only the final result to **both** the customer and the merchant's website.

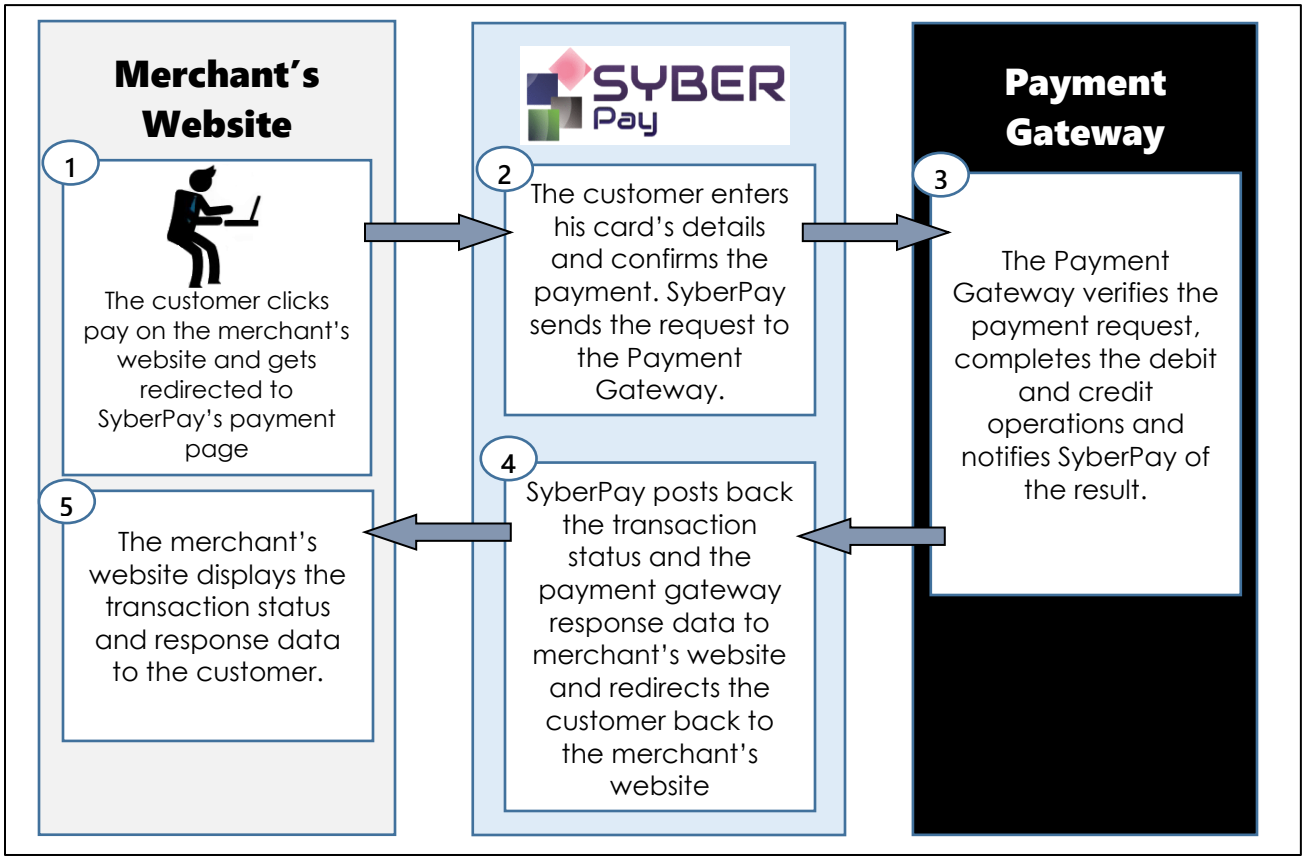

Figure 1: Transaction Flow Overview

3. Integration with SyberPay

3.1. Overview

Integration with SyberPay is performed using its HTTP API and employing JSON as its data interchange format. To be more precise it uses HTTPs to provide end-to-end security for all the messages exchanged. All calls to the API should be made to a URL following this format:

https://[SYBERPAY_URL]/[SERVICE_PATH]

Whereas:

SYBERPAY_URL: is the URL used by SyberPay following the standard web links format. Different SyberPay environments (e.g. production and test) have different URLs but all follow the same format.

SERVICE_PATH: This has a one-to-one mapping to the web services provided by SyberPay. Web services here correspond to the functions made available by SyberPay through its API. Table 1 lists the web services provided by SyberPay with a brief description for each.

Table 1: List of SyberPay Services

Service name	Service description	SERVICE_PATH
Payment	Request to initiate a payment operation for the Merchant.	getUrl
Payment Status	Retrieves the status and data of a particular transaction.	payment_status

3.2. SyberPay Web Services

As summarized in the table, SyberPay provides two main web services. This section outlines each web service and discusses its use case in more details. These web services are called using HTTP POST to the service URL while the service parameters are encoded as JSON and transferred in the payload of the request.

3.2.1. Payment

This web service is requested by calling a URL as:

https://[SYBERPAY_URI]/getUrl

It indicates that a customer at the merchant website have chosen to pay online using SyberPay. As the service path name indicates, this service returns a URL to which the customer should be redirected. This is the URL where the customer enters the card details to be used for payment. After processing the payment transaction, SyberPay informs the merchant website that the transaction is complete.

As per this flow the customer enters the financial data at SyberPay's website not the merchant's. SyberPay platform is designed as such so that all responsibilities of handling the customers' data is transferred from the merchant side and is handled instead by SyberPay. This way the merchant doesn't have to get certified with any authority for handling such sensitive information.

3.2.2. Payment Status

This web service is requested by calling a URL as:

https://[SYBERPAY_URI]/payment_status

This service is used to get the detailed information of a particular transaction. At the end of the Payment service, SyberPay notifies the merchant website that a particular transaction is now complete and its result is ready. The merchant website uses this service to get detailed results of that transaction and to

inform the customer of that outcome. More details on this in section 5.4 Instant Payment Notification (IPN).

3.3. Merchant Registration

In order to integrate with SyberPay, the merchant must register itself with SyberPay first. It is at this registration step that the merchant is all of his required identification data. The main Id for the merchant is the application ID which is a unique ID generated by SyberPay and uniquely identifies each merchant. Also during the registration the list of services allowed for this merchant is defined. Each service has a unique ID (known adequately as Service ID) to differentiate between the services offered by the same merchant. As an example, a telecom operator can offer two main services at its website; one service is toping up prepaid lines and another is paying the bills of postpaid lines. Differentiating between the services allows for better controls over the details of the transactions such as having different fees for each service.

4. Web Service Requirements

The previous sections described an overall view of the transaction and the web services used by SyberPay's Merchants. Now this section details how these services are called; i.e. it answers the questions of which parameters to use and what parameters to expect in the responses**?**

4.1. Payment

Table 2 & Table 3 list the fields of the request and the response of this service with a brief description for each fields and its format. All of the fields are mandatory except for the *payeeId* as SyberPay will use the value registered with the merchant by default if this field was not present in the message.

Table 2: Payment Service Request Fields

Field Name	*Field Description*
applicationId	*Unique ID assigned to the merchant application by SyberPay*
payeeId	*Unique ID for the merchant as provided by Payment Gateway*
serviceId	*Unique ID for the merchant service provided by SyberPay*
customerRef	*Transaction reference ID generated by the merchant application*
amount	*Total amount of the transaction*
currency	*Currency code of the transaction*
paymentInfo	*Payment information as a key/value map*
hash	*Hash string to ensure integrity of the message*

The first three fields (i.e. the three IDs) are used to identify the exact service requested by the customer which are to be paid for. The *applicationId* is constant per merchant while the *payeeId* & *serviceId* are fixed per service. The customer reference field is the ID used by the merchant to identify that particular transaction. Having it sent to SyberPay eases the cross comparison between the two entities at any point of time. The amount and currency are self-explanatory. The Payment information field (*paymentInfo*) is any additional information required to fulfil the requested service (e.g. phone number in case of postpaid bill payment & meter number in case of electricity purchase). The final field, hash, is generated to ensure the integrity of the message and guard against man-in-the-middle attacks.

Table 3: Payment Service Response Fields

Field Name	Field Description
applicationId	*Unique ID assigned to the merchant application by SyberPay*
payeeId	*Unique ID for the merchant as provided by Payment Gateway*
serviceId	*Unique ID for the merchant service provided by SyberPay*
customerRef	*Transaction reference ID generated by the merchant application*
amount	*Total amount of the transaction*
currency	*Currency code of the transaction*
paymentInfo	*Payment information as a key/value map*
tranTimestamp	*The timestamp of the transaction*
responseCode	*Code representing the status of the transaction*
responseMessage	*Textual Message describing the status of the transaction*
paymentUrl	*Payment URL generated by SyberPay for this transaction*

All of the fields from the request are echoed back in the response, plus four more fields; *transTimestamp* is the precise timestamp of this transaction at the time of its processing at SyberPay. Both the *responseCode* and *responseMessage* convey SyberPay readiness to process the payment transaction as initiated by the request. In case of any issue (e.g. wrong *applicationId* or a hash validation error) these two fields will reflect the situation and allow the merchant website to recover accordingly. Finally, the *paymentURL* is the URL to which the customer should be redirected to perform the payment operation. This last field is only returned in case the request has passed SyberPay's validation process as indicated by a *respondeCode* of '1'.

4.2. Payment Status

This web service is used to retrieve the status and data of a given payment transaction. It is called by the merchant's site once notified about the transaction completion by SyberPay.

Table 4: Payment Status Service Request Fields

Field Name	Field Description
applicationId	*Unique ID assigned to the merchant application by SyberPay*
transactionId	*Unique ID of a payment transaction in SyberPay*
hash	*Hash string to ensure integrity of the message*

The *applicationId* and hash fields serve the same purpose as the fields sent for the payment service request. The *transactionId* field contains a unique payment transaction id that the merchant is inquiring about its status. More details about the payment process is outlined in section 5 .

Table 5: Payment Status Service Response Fields

Field Name	Field Description
applicationId	*Unique ID assigned to the merchant application by SyberPay*
transactionId	*Unique ID of a payment transaction in SyberPay*
tranTimestamp	*The timestamp of the transaction*
responseCode	*Code representing the status of the transaction*
responseMessage	*Textual Message describing the status of the transaction*
payment	*All the information regarding a given payment transaction, returned as an Object.*

status	*The status of the service response*

The new fields here are the *payment* and the *status*. The *responseCode* and *responseMessage* fields again serve the same purpose of indicating that SyberPay has accepted the previous request and that request has passed its validation. The *status* field is also similar to the *responseMessage* conveying SyberPay's response to the request but only contains the status of the transaction as one of two values; "Success" or "Invalid". The most important field here is the *payment* which contains all of the details of the transaction identified by *transactionId*. Its fields' composition is very dynamic but in essence it contains two sets of fields; first, all the fields sent by/to the merchant itself in the payment service response (i.e. the fields of Table 2) but with the *responseCode* and *responseMessage* filled with the values provided by the payment gateway. Note that in the original payment service response these two fields contain the responses from SybePay and indicate just that the initial payment service request has passed the validation. In the new context (in the *payment* fields) they contain the response to the actual payment operation as performed by the payment gateway. The second set of fields contains a single field named *responseData* which contains a set of fields passed along from the payment gateway for the customer's benefit. A simple example for the data contained in this field would be the postpaid line bill information for a bill inquiry transaction.

5. Payment Process

This section describes the flow of messages and web services' calls to be followed in order to perform an online payment through SyberPay. Figure 2 shows the details of that flow.

5.1. Merchant's Registration

As mentioned before, first the merchant must register with SyberPay in order to be identified by the platform. At this registration step the following items are generated:

1. Application ID: This is the unique identifier of that merchant in SyberPay platform. Each merchant at registration receives a single application ID that is used in all messages and calls as the field *applicationId*.
2. Payee ID: This item is sent as the field *payeeId* and it determines who gets the credit part of the payment transaction; e.g. If a merchant's website is offering phone bill payments then the phone operator should be getting the credit while the merchant is just an intermediate between the customer (the one paying for the service and who will debited for it) and the phone operator (the payee who will be credited for the service). In this scenario the payee ID will be that of the phone operator. In a different scenario, if the merchant is offering a service for which he/she is getting the credit part such as an online ecommerce website, the payee ID should be that of the merchant himself. This ID is generated at the payment gateway and handled by SyberPay for its merchants.
3. Service ID: Each service provided by any payee has its own unique ID. This allows for much finer control over handling of the services; e.g. different fees distribution for each service. This is transported as the field *servcieId* during message exchange.
4. Application Key & Application Salt: Two secrets that are never exchanged in the clear and are only used for generating hashes (i.e. the field *hash* in the messages). These fields are generated by SyberPay and are shared with the merchant who should keep them secret and never share them with any 3rd party. The exposure of these two secrets will compromise the integrity of the

messages exchanged between the merchant and SyberPay and should be communicated immediately to void the compromised keys and generate new ones.

5. Notify, Return and Cancel URLs: These are the URLs used to redirect the customer back to the merchant website during step 6 (or step 4 in case of cancelation) of Figure 2. More on that in the next section.

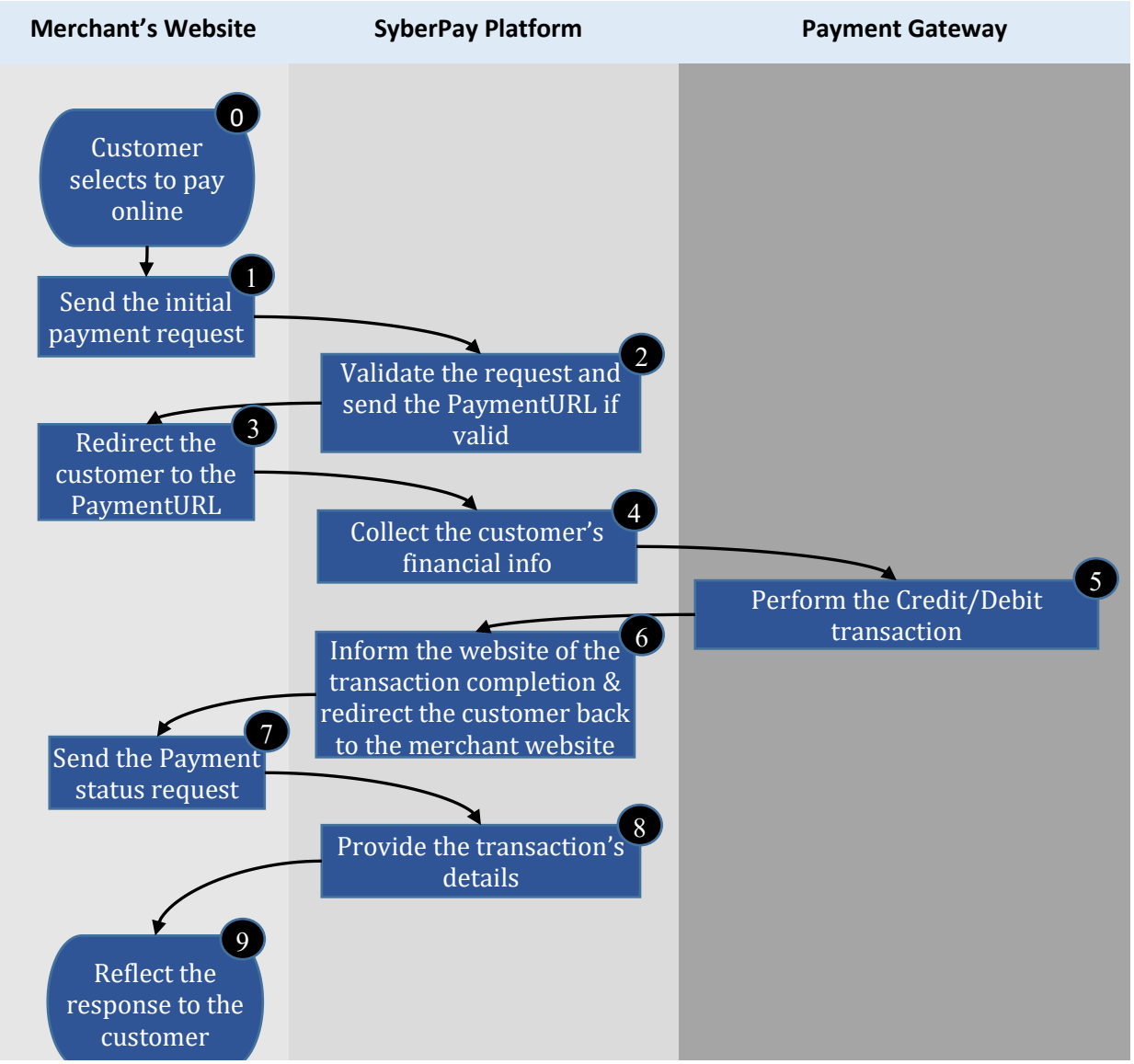

Figure 2: Payment Process Detailed

5.2. Payment Process Detailed

Figure 2 shows the detailed payment process as all validations step succeeds and the payment gateway's processes the transaction successfully. Following the same numbering scheme as with the figure, this is further explained as follows:

0. The customer has been browsing the merchant's website and has decided to pay for a particular service using SyberPay as the payment method.

1. The merchant's website prepares the fields listed in Table 2, parses them in JSON and sends (more precisely POSTs) this message to the payment service URL.
2. SyberPay receives the payment request, validates the hash and the rest of the fields and replies with the JSON payload of the fields in Table 3. Note that if the validates succeed, the response includes the URL to which the customer is to be redirected to complete the payment process. This URL is generated on the fly in response to the merchant's payment request and will expire after 10 minutes.
3. The merchant's website redirects the customer to the Payment's URL.
4. The customer is presented with the page showing the amount and currency to be paid and asking the customer to fill his/her financial details (i.e. Card PAN, expiry date and iPIN). After the customer's confirmation, SyberPay instructs the payment gateway to perform the payment transaction.
5. The payment gateway fulfills the debit and credit parts of the transaction and informs SyberPay of the result. Note that for some transactions the credit part may not be performed at the same time with the debit and may be delayed, but the debit transaction itself is never delayed; i.e. the customer's account is always checked and the requested amount debited from it but crediting the merchant's account may be delayed.
6. Upon receiving the result of the transaction from the payment gateway, SyberPay informs the merchant's website of the completion of the transaction by doing an HTTP POST to the notify URL (one of the URLs defined during the registration phase). As the POST parameters, SyberPay sends three fields; the transaction ID, a random token and a hash. The details are mentioned in section 5.4. Then SyberPay redirects the customer to the Return URL (another URL from the registration phase).
7. Now the merchant's website knows there is a completed transaction at SyberPay side. To get its details, the website forms the JSON payload of the fields in Table 4 and sends the payment status request to SyberPay. Note these are the same fields received from SyberPay's end at step 6.
8. If the request passes the validation, SyberPay sends the payment status response as composed of the fields of Table 5. This includes all of the details of the response from the payment gateway.
9. Finally the merchant's website uses the response received to update its data and reflect the results to the customer.

Two of the most interesting and common cases are the validation failure of the payment service request and the customer cancelling the payment after getting redirected to SyberPay *PaymentURL*. For the 1st case, SyberPay will not return the *PaymentURL* and the *responseCode* and *responseMessage* will reflect the reason behind the validation failure (e.g. wrong *serviceId*). Thus the merchant's website should be able to recover from this situation and either fix the cause of the error or present a suitable error message to the customer. The latter case – the cancellation – the customer is simply returned to the Cancel URL; i.e. the remaining URL of the three defined during the registration phase. Again, this allows the merchant's website to recover accordingly (e.g. preserve the customer's checked out items).

5.3. Hash Calculations

During the exchange of the various messages between the merchant and SyberPay, hash values need to be calculated and transmitted for a number of scenarios to allow the receiving party to check the integrity of the message. All of these hash calculations are based on the SHA256 algorithm variation. These calculations are detailed next.

- For the payment request call made from the merchant's website to SyberPay, the hash is calculated as follows:

hash = SHA-256 (applicationKey | applicationId | serviceId | amount | currency | customerRef | applicationSalt)

- For the notification sent to the notify URL sent from SyberPay to the merchant's website, the hash is calculated as follows:

hash = SHA-256 (applicationKey | transactionId | token | applicationSalt).

- For the payment status request sent in response to the previous notification, the hash is calculated as follows:

hash = SHA-256 (applicationKey | applicationId | transactionId | applicationSalt)

5.4. Instant Payment Notification (IPN)

This is the name of the mechanism by which SyberPay is able to return the results of a transaction to the merchant at the instant it becomes available without requiring the merchant to have an SSL certificate at the website and without compromising the security of the payment process.

As the name suggests this is based on notifying the merchant's website of the results. But instead of sending the results of the transaction, it merely notifies the merchant's website that a particular transaction as identified by its *transactionId* is finished. Then the merchant's website queries SyberPay about the details of the transaction's results in a separated web service call (i.e. payment status request call). Finally, SyberPay responds by providing the full details of that result in the payment status response.

That is the reason why the merchant should provide three URLs during the registration step: **notify**, **return**, and **cancel**. As mentioned previously, the notify URL is used to notify the merchant's site that the payment transaction has completed. The return URL is used to redirect the customer back to the merchant's site after it is notified. The cancel URL is used if the payment process is cancelled by the customer after he was redirected to SyberPay for payment. For the notification, SyberPay uses HTTP POST to send the following fields to the merchant's notify URL:

1. Transaction ID: The unique Id of the transaction which just completed and this merchant should request its status. Note this is the unique ID used by SyberPay platform and is not the merchant's locally significant ID (i.e. *customerRef*)
2. Token: A random string that is used in the hash calculation and sent also in the POST variables to allow the merchant's website to verify that this message has indeed originated from SyberPay and to verify its integrity.
3. Hash: The actual hash value as calculated as shown previously in section 5.3.

As per the flow previously discussed, once SyberPay notifies the merchant's website through HTTP POST and the notify URL, it redirects the customer to the return URL. In this redirection, SyberPay also uses

HTTP POST to send the *customerRef* (i.e. the merchant's unique ID for the transactions) to allow the merchant's website to identify the transaction and customer being redirected. Once the merchant's website is notified through its notify URL, it should initiate a payment status request and retrieve the status of that transaction to complete the cycle and inform the customer of the result and update its data accordingly.

6. Appendix A: Acronyms & Definitions

API	Application Programming Interface
CBoS	Central Bank of Sudan
Customer	The end user paying for the service from the merchant's website
HTTP	Hypertext Transfer Protocol
iPIN	Internet Personal Identification Number
IPN	Instant Payment Notification
JSON	Java Script Object Notation
Merchant	The entity offering services online
MIME	Multipurpose Internet Mail Extension
PAN	primary account number
Payment Aggregator	Provides a single point of contact for several payment options possibly through different payments gateway.
Payment Gateway	Provides that actual ePayment service of debiting and crediting the responsible parties.
SHA	Secure Hash Algorithm
SSL	Secure Socket Layer
URL	Uniform Resource Locator
UUID	Universally Unique Identifier

7. Appendix B: List of SyberPay Fields

Field name	*Field Description*	*Format*
amount	***The total amount of the transaction***	***Number(10,2)***
applicationId	***The unique ID assigned to the merchant application by SyberPay.***	***String(10)***

currency	*The currency code of the transaction (ex. "SDG")*	*String(3)*
customerRef	*The transaction reference ID as generated by the merchant's website/application.*	*String(36)*
hash	*The hash string generated using a hash algorithm (SHA-256) to ensure integrity of the message*	*String(64)*
payeeId	*The unique ID for the payee as provided by payment gateway*	*String(10)*
payment	*All the information regarding a given payment transaction, returned as an Object.*	*JSON Object*
paymentInfo	*The payment information as a key/value map (ex. "AccountNo": "3648", "CustomerName": "Name")*	*JSON object*
paymentUrl	*The payment page URL generated by SyberPay for this transaction.*	*String(200)*
responseCode	*Code representing the status of the transaction (see the related Response Codes document)*	*Number(3)*
responseData	*Any response data that is returned from the payment gateway application.*	*JSON Object*
responseMessage	*Message describing the status of the transaction (see the related Response Codes document)*	*String (100)*
serviceId	*The unique ID for the merchant service provided by SyberPay*	*String(15)*
status	*The status of the service response*	*String (100)*
transactionId	*The unique identifier of a payment transaction in SyberPay.*	*String(36)*
tranTimestamp	*The time stamp of the transaction*	*timestamp*

Where:

- Number(n,m): A real number having an integer part of max length n digits and decimal part of m digits.
- Number(n): an integer of max length of n digits
- String(N): A series of ASCII characters with max length of N.
- timestamp: A string representing the date and time information formatted as "YYYY-MM-DD,HH:MM:SS" (e.g. "2015-12-13,14:04:58")
- JSON Object: A set of key & value pairs encoded as JSON.

8. Appendix C: Sample Request/Response Pairs

8.1. Payment web service

8.1.1. Request Format


```
{
"applicationId":"0010010001",
"payeeId":"0010010009",
```

"serviceId":"001001000901",
"customerRef":"628db23f",
"currency":"SDG",
"amount":100.0,
"paymentInfo":
{
"accountNo":"123456",
"customerName":"Mohammed Ahmed",
"serviceType":"123"
},
“hash”: “e1280edbe5bf00c237e6b724d63f0ddab2e48b1ee3b5db7573b59f293cae655a”
}

8.1.2. Response Format

{
"applicationId":"0010010001",
"payeeId":"0010010009",
"serviceId":"001001000901",
"amount":100.0,
"currency":"SDG",
"customerRef":"628db23f",
"tranTimestamp":"2015-08-13,10:04:58",
"paymentInfo":
{
"accountNo":"123456",
"customerName":"Mohammed Ahmed",
"serviceType":"123",
},
"responseCode":1,
"responseMessage":"Approved",
"paymentUrl":"https://www.sybertechnology.com/syberpay/payment/1593135f-4110-4edc-bbd0-ebe134968498"
}

8.2. Payment Status Web Service

8.2.1. Request Format

{
"applicationId":"0010010001",
"transactionId":"1593135f-4110-4edc-bbd0-ebe134968498",
“hash”: “e1280edbe5bf00c237e6b724d63f0ddab2e48b1ee3b5db7573b59f293cae655a”
}

8.2.2. Response Format

{
“status” : “Successful”,
"responseCode":1,
"responseMessage":"Approved",

```
“payment” : {
        "applicationId":"0010010001",
        "payeeId":"0010010009",
        "serviceId":"001001000901",
        "amount":100.0,
        "currency":"SDG",
        "customerRef":"628db23f",
        "tranTimestamp":"2015-08-13,10:04:58",
        "paymentInfo": {
            "accountNo":"123456",
            "customerName":"Mohammed Ahmed",
            "serviceType":"123",
        },
        “responseData” : {
                “receiptNo” : “972XY30”
        },
        "responseCode": 0,
        "responseMessage": "Approved",
        "status": “Successful”
}
}
```


9. Appendix D: SyberPay Response Codes

Code	Message	When this code is generated
1	Approved	All is good and well
2	Invalid Transaction Amount/Currency	Either the amount and/or currency are not supported
3	Invalid Service Id	The serviceId is not available to this merchant
4	Invalid Request Format	The format used in the request fields in not supported
10	Invalid payment Information	The payment information fields are not supported
11	System Error	An internal error to the SyberPay Platform
12	Invalid Customer Reference	The *customerRef* is unknown to SyberPay platform
13	Invalid Payee	The *payeeId* used in not supported
14	Invalid Application	The *applicationId* used in not supported
15	Authentication Failed	The message integrity could not be verified
16	Connection Failed	Connection to payment gateway failed
17	Connection Timed Out	Connection to payment gateway timed out
18	Invalid Input Entry	PAN#, expiry date, iPIN
19	Invalid Transaction Id	The *transactionId* used is not available for this merchant

For each responseCode in the table above the responseMessage field will contain the corresponding message text.